

INDEPENDENT PROOF PROGRAMME

Erdos Problem #742

Daily adversarial audit

Mathematical findings, evidentiary status, repository integrity and session utilisation for the 24 hours ending 22 September 2026.

Audited predecessor	0d11049574fce97f4630a4b71e8da78edbb7836c
Pivot commit	5825c203da78ecaefaa63568f0883de25518df9f
Focused sessions	1 of 24 before the go/no-go decision
Primary result	Candidate threshold improved from 7/12 to 250/429
Audit classification	Internal adversarial review; not external verification

Verdict

Continue. The first focused session produced a real candidate threshold improvement, but the graph-to-profile bridge and equality geometry remain the load-bearing gates.

Prepared for a mathematically sophisticated external reader. All theorem labels below retain the repository's candidate/internal-review trust boundary unless stated otherwise.

1. Executive assessment

The claimed all-order implication $n \geq 6$ and $\Delta(G) \geq 250n/429$ implies $e(G) < \text{floor}(n^2/4)$ survives this audit at candidate level. The audit replayed the rational scalar certificate, convex large- a assembly, all eight finite exceptional rows, the canonical 7/12 checker and the threshold-capacity algebra. Every executed check passed.

The improvement is numerically modest but mathematically material: $250/429 = 0.582750\dots$ is strictly below $7/12 = 0.583333\dots$. The live strip is now $n/2 < \Delta < 250n/429$.

250/429 threshold	Retained as internally checked candidate theorem; external review open.
Graph-to-profile spine	No flaw found in targeted replay; remains the principal shared proof risk.
External-proof independence	No use of the Erdos742 $e+\text{disj}+X$ core found in source/dependency search.
Full-Hall plateau	Valid asymptotic relaxation obstruction; not an actual graph.
High-codegree corollary	Arithmetic survives, but theorem scope was overbroad and has been corrected.
Equality characterization	Open in the remaining strip; odd balanced complete bipartite graphs are a mandatory control.

The 24-session gate

Focused-session count is **1/24**. The gate is not due. Session 1 already satisfies its minimum progress criterion by producing a strict threshold improvement and a concrete realizability obstruction. That supports continuation, but the explicit session-24 reassessment remains binding.

2. Threshold proof audit

The proof starts with $a=n-1-b$, $b=\Delta(G)$, and $t=e(G)-b(n-b)$. The inherited selected/residual profile lemma yields $r \geq \sum \Phi(s_i/a) - a/4$ and $2t \leq S-r$. The new rational optimization proves $f(x)=x-\Phi(x) < 121/1569$.

$$t < 121 \ a^2 / 3138 + a/8$$

For $b \geq 250n/429$, the continuous degree margin is positive from $a=65$ onward. Exact assembly over $2 \leq a \leq 64$ leaves $a=\{4,6,9,11,14,19,24,29\}$; each is closed by the preserved finite threshold-capacity certificate. The $a=0$ star and $a=1$ edgeless- F cases are separate and correct.

The strictness and floor terms were checked. Larger eligible b only increases the required surplus, so testing the least eligible degree is sufficient. The diagnostic degree regression checked 233,515 pairs below 7/12 through $n=10,000$; it is consistency evidence, not the proof.

Checker	Result	Interpretation
check_39_67.py	PASS	Exact rational certificate, threshold ladder, eight finite exceptions
check_threshold_algebra.py	PASS	177,120 integer threshold triples
check_degree_regression.py	PASS	233,515 degree pairs through $n=10,000$
src/check_7_12.py	PASS	Inherited scalar, finite certificates and 5,207,079-pair regression
four plateau/rigidity checkers	PASS	Arithmetic and margin checks only

3. Graph bridge, plateau and realizability

The audit found no blocking quantifier or direction error in the complement/quasi-edge ledger, source-demand injection, supplement dichotomy, threshold-capacity count or Cauchy projection. The supplement step is delicate: a supplement-label edge need not be residual, and the proof works only because the selected alternative invokes the already-proved source-demand inequality.

This remains same-project internal replay. The next session should reconstruct the complete bridge independently from raw D2C criticality and then exercise it on actual graphs under multiple legal selections.

The demand-1/4, $b/a=139/100$ plateau is a useful negative result. It survives scalar capacity, full selected-incidence Hall, residual graphicality, orientation balance, pair uniqueness and disjoint selected/residual matrix margins. It is explicitly an aggregated asymptotic relaxation, not a D2C graph. It shows that another one-dimensional profile inequality is not a credible path to 1/2.

Scope defect found and repaired

Original statement	Listed $Q \geq a^2/4$ and upper degree/endpoint caps, but omitted the lower degree and selected-endpoint hypotheses needed by robust stability.
Conflict	The robust average bound uses $Q \leq a^2/4$, while the convexity lower bound on pair occurrences uses $Q \geq a^2/4$.
Repair	State the full joint plateau hypotheses, which force $Q=a^2/4$ in the exact normalization. Arithmetic and the exact plateau conclusion survive; the result is not a general band theorem.

4. Independence and equality geometry

A repository-wide source search found no use of the external Erdos742/Erdos742 universal $e+disj+X$ inequality in either the new strip package or the inherited 7/12 profile proof. The new route uses complement/quasi-edge selection and selected/residual profile capacity. On the available evidence it is structurally independent. This does not replace an external comparison of proof cores before any novelty claim.

The equality issue is not cosmetic. For even n , $K_{(n/2, n/2)}$ sits at $\Delta=n/2$. For odd n , $K_{((n-1)/2, (n+1)/2)}$ has $\Delta=(n+1)/2$ and lies inside the open strip. The signature plateau has positive selected demand and explicitly misses this zero-demand boundary. A future closure must isolate or stably recover the complete-bipartite geometry.

5. Repository and process audit

The audited head failed the protected status guard because its final telemetry-only commit left the CURRENT-STATUS block unchanged. README also contradicted its correct top-level pivot statement in later sections that still called eventual-D2C the active chain. Both defects were repaired, while the historical failure and earlier eventual-D2C materials remain preserved.

Before the pivot, the day produced extensive Q3/eventual-D2C work. It is not a dependency of the 250/429 theorem and was not promoted by this audit. Same-day invalidations and repairs remain visible, including the asymmetric scanner bug, a falsified single-class guess, matching/cancellation failures and the corrected odd-u gap formula.

6. Session utilisation

There were 23 scheduled forward slots. Fourteen have some telemetry and nine have none. Five have fully verified whole-slot records; four met 50 minutes and one recorded 48 minutes. Closed non-overlapping intervals establish at least 371m25s of forward work and 195 substantive units. This is a lower bound, not a whole-day utilisation estimate.

Slots 01-13

01	01:00:26	initial PENDING; cont. 01:48:32	PENDING / 01:50 / 01:59	UNV / >=20:59	27	UNVERIFIED
02	MISSING	MISSING	MISSING	UNVERIFIED	-	MISSING
03	03:00:51	initial PENDING; late 03:59:26	04:00:45 late seg.	UNVERIFIED	3	UNVERIFIED
04	MISSING	MISSING	MISSING	UNVERIFIED	-	MISSING
05	05:00:47	initial PENDING; late 05:59:12	06:00:01 late seg.	UNVERIFIED	4	UNVERIFIED
06	MISSING	MISSING	MISSING	UNVERIFIED	-	MISSING
07	07:02:17	PENDING	PENDING	UNVERIFIED	0	STALE
08	08:04:14	initial PENDING; cont. 08:47:56	08:51:26	UNV / >=16:23	20	UNVERIFIED
09	09:00:41	initial PENDING; cont. 09:47:52	09:59:10	UNV / >=19:37	18	NONCOMPLIANT
10	MISSING	MISSING	MISSING	UNVERIFIED	-	MISSING
11	11:00:26	initial PENDING; cont. 11:48:30	11:59:20	UNV / >=19:59	17	UNVERIFIED
12	MISSING	MISSING	MISSING	UNVERIFIED	-	MISSING
13	13:01:30	initial PENDING; cont. 13:49:16	13:51:15	UNV / >=18:58	12	UNVERIFIED

UNV means unverified. Multiple records naming the same trigger were reconciled as segments of one scheduled slot; missing boundaries were not inferred.

6. Session utilisation - continued

Slots 14-23

14	14:01:25	14:55:57	14:57:52	57:17 / >=22:10	15	UNVERIFIED
15	15:02:03	15:55:44	15:59:30	57:27 / 52:58	26	MET
16	MISSING	MISSING	MISSING	UNVERIFIED	-	MISSING
17	17:03:31	17:52:05	17:53:22	49:51 / 48:00	10	NOT MET
18	18:00:38	18:51:50	18:52:55	52:17 / 50:09	15	MET
19	19:01:53	19:53:30	19:54:27	52:34 / 50:51	16	MET
20	20:00:28	20:52:59	20:55:43	55:15 / 51:21	12	MET
21	MISSING	MISSING	MISSING	UNVERIFIED	-	MISSING
22	MISSING	MISSING	MISSING	UNVERIFIED	-	MISSING
23	MISSING	MISSING	MISSING	UNVERIFIED	-	MISSING

Scheduled slots	23
Any telemetry	14
No telemetry	9
Fully verified whole slots	5
Verified target met / not met	4 / 1
Evidenced forward lower bound	371m25s
Recorded substantive units	195
Whole-day utilisation	UNVERIFIED

7. Prioritized next 24 hours

- Reconstruct the complete selected-quasi-edge/profile lemma independently from raw D2C criticality. Include selection existence, pair uniqueness, the source-demand injection, the supplement dichotomy and every threshold-capacity quantifier.
- Build an actual-graph regression that constructs A, B, F, selected/residual incidences, s_i and z_h under multiple legal selections. Mandatory controls: balanced complete bipartite graphs of both parities and X_3 .
- Only after those gates pass, attack the corrected high-codegree cluster from raw criticality. Seek either a quantitative upper bound below $0.00351 a^2$ or a bounded cluster/blow-up classification.
- Develop a separate zero- and near-zero-demand equality branch whose limiting case is visibly the balanced complete bipartite graph.
- Maintain the one-session-per-hour count and apply the evidence-based continue/pivot decision at the first audit after session 24.

Stop / pivot criteria

- Any graph-to-profile flaw freezes every dependent threshold claim.
- Any high-codegree route using only scalar or Hall marginals has returned to the certified plateau barrier.
- Bounded failure to find a graph is not a nonrealizability theorem.
- Any strip closure excluding odd balanced complete bipartite graphs has mishandled equality.

8. Evidence map

Artifact	Repository path
Threshold and scalar obstruction	project/research/general_n/2026-09-21-independent-742-strip-v1/FIRST_OBSTRUCTION_AND_39_67.md
Graph-to-profile hostile replay	project/research/general_n/2026-09-21-independent-742-strip-v1/GRAPH_TO_THRESHOLD_HOSTILE_REPLAY.md
Full-Hall plateau	project/research/general_n/2026-09-21-independent-742-strip-v1/FULL_HALL_PLATEAU_OBSTRUCTION.md
Signature union rigidity	project/research/general_n/2026-09-21-independent-742-strip-v1/SIGNATURE_UNION_RIGIDITY.md
Robust stability	project/research/general_n/2026-09-21-independent-742-strip-v1/ROBUST_SIGNATURE_STABILITY.md
Corrected density corollary	project/research/general_n/2026-09-21-independent-742-strip-v1/POSITIVE_DENSITY_CODEGREE_CLUSTER.md
Full audit note	project/research/general_n/2026-09-22-daily-red-team-audit-v1/DAILY_RED_TEAM_AUDIT.md

Commit anchors

Pivot 5825c203da78... | strip package a69487d826eb... | focused-session checkpoint fd9f816c84e6... | audited head 0d11049574fc...

Final status: meaningful candidate progress, one repaired scope defect, no complete proof, and a sharply defined next verification gate.